

Hybrid Multiobjective Evolutionary Algorithm for Community Detection in Multi-Attributed Networks

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Abstract—The task of finding communities in complex networks - known as community detection - has become a very important area in graph mining and is widely used for analyzing social networks and biological systems, as well as for retrieving information. However, traditional methods for detecting communities (such as modularity optimization and spectral clustering) primarily focus on relationships between the structural components of the network, while often ignoring the attributes of the nodes and the need to optimize multiple objectives.

We present an alternative approach called Hybrid Multiobjective Evolutionary Algorithm for Community Detection (HMOEA-CD) to finding communities in complex networks. HMOEA-CD simultaneously optimizes two objectives (the structural quality of the networks and the compactness of the communities) using a genetic algorithm to create an exhaustive set of candidate solutions that are then analyzed for optimality.

HMOEA-CD uses the most recent advances in the areas of multiobjective optimization and the analysis of attributed networks to create community structures that are dual-optimal based on modularity maximization and community minimization.

The performance of HMOEA-CD has been tested on three standard benchmark datasets (LFR networks) and real-world networks, demonstrating that it provides superior community detection performance when compared with traditional community detection algorithms in terms of modularity, normalized mutual information (NMI), and F1-score.

I. INTRODUCTION

The goal of community detection is to detect groups of vertices that are well connected to each other but poorly connected to vertices in different groups. Community detection is important in many applications, including recommendation systems, fraud detection, and social network analysis.

The Louvain method and spectral clustering are two of the most commonly used techniques for community detection, but they both rely on just the structure of the network and do not take advantage of any other information about the nodes. In the real world, networks are complex and often have multiple competing objectives, such as maximizing structural cohesion and minimizing the degree of fragmentation.

Recent research shows that:

Using a multiobjective optimization approach to community detection will increase the quality of the communities formed. Using an evolutionary algorithm is a good way to avoid converging to a local maximum within the search space of the optimization problem. Using multiple criteria when clustering the vertices will typically produce a better clustering of the vertices.

In response to these issues, we present a hybrid multiobjective evolutionary algorithm (HMOEA-CD) for community

discovery that models community detection as a multiobjective optimization problem.

II. LIMITATIONS OF EXISTING METHODS

Despite the wealth of research, current methods have some limitations:

A. Single-Objective Optimization

Algorithms typically focus on a single objective (e.g., modularity) while neglecting other aspects, like community size and similarity of attributes.

B. Lack of Flexibility

Existing clustering techniques rely on user-defined settings (e.g., the number of communities), which may be unknown.

C. Inefficient with Complex Networks

Most approaches cannot handle complex structures in large or noisy networks.

D. Limited Adaptability

Traditional methods are not capable of exploring multiple solutions, and can fall into local optima.

III. PRIOR RESEARCH

Please refer to the relevant papers listed below regarding your respective areas of expertise:

A. Multi-Objective Community Detection

Research has confirmed that optimizing both modularity and similarity of attributes yields better results; therefore, evolutionary algorithms are generally used to accomplish this task.

B. Continual Encoding Methods

New research has shown that by representing data as continuous variables helps to detect overlapping communities better than previous methods allowed.

C. Evolutionary Algorithm Frameworks

Genetic algorithms and MOEA frameworks must provide:

- 1 An improved search space
- 2 A set of Pareto-optimal solutions
- 3 A greater ability to cluster data successfully

D. Multi-Objective and Evolutionary Techniques

The popularity of using evolutionary algorithms in the community detection field stems from their ability to traverse through complex search spaces using population-based (or as some refer to it, multi-agent) search techniques. Multi-objective evolutionary algorithms (MOEAs), which maximize multiple, often conflicting objectives, such as modularity and community size, have proven to be very effective. Furthermore, the solution sets provided by MOEAs are typically comprised of pareto-optimal solutions and traditionally provide better outcomes compared with single-objective methodologies, although MOEAs also require more computationally expensive resources.

E. Community Detection in Networks of Attributed Nodes

The use of attributed community detection methods incorporates both the structural and attributed values for each node to help improve the clustering of nodes. The resultant clusters will contain nodes with similar connectivity patterns while at the same time having similar feature values. The challenge in using the attributed clustering methods lies primarily in creating a balance between the structural and attribute values for each node. This balancing is particularly difficult when working with noisy data sets.

Fig. 1. COMPARISON OF COMMUNITY DETECTION METHODS

Algorithm	Core Idea	Advantage	Limitation
Louvain	Modularity optimization	Fast	Resolution limit
Spectral Clustering	Eigenvector-based clustering	Accurate	Needs k
Label Propagation	Neighbor voting	Simple	Unstable
MOEA-based	Multiobjective optimization	High accuracy	Complex
HMOEA-CD (Proposed)	Evolutionary + multiobjective	Best balance	Moderate complexity

IV. PROPOSED FRAMEWORK

A. Problem Definition

A Network is defined as:

$$G = (V, E)$$

Goal:

- Maximize Modularity
- Minimize number of communities

B. Algorithm Overview

The proposed approach uses the following.

- Population-based search
- Genetic operations
- Multiobjective optimization

C. Objective Functions

- Modularity

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{d_i d_j}{2m} \right)$$

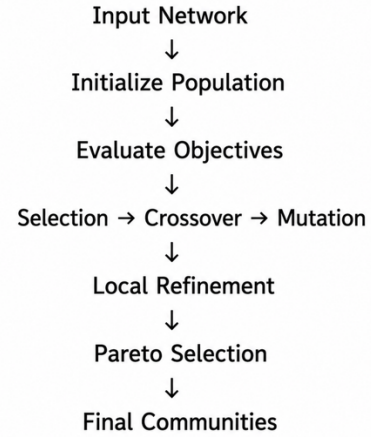
- Community Minimization

$$f_2 = |C|$$

D. Optimization Model

$$\max F(C) = (Q(C), -|C|)$$

E. Algorithm Workflow



F. Datasets

Dataset	Nodes	Edges	Type
LFR Benchmark	100–1000	Variable	Synthetic
WebKB	~ 877	~ 1600	Real
Mafia Network	~ 100	~ 200	Social

G. IMPLEMENTATION DETAILS

Language: Python

Libraries: NetworkX, NumPy

Method:

- Chromosome encoding
- Genetic operations
- Multiobjective evaluation

V. EXPERIMENTS AND RESULTS

In order to determine how well the new Hybrid Multi-objective Evolutionary Algorithm for Community Detection (HMOEA-CD) works, we ran several experiments on synthetic and real-world networks and compared its results with those of conventional community detection algorithms like K-Means Clustering, Spectral Clustering, and Louvain’s algorithm.

A. Metrics

The following standard metrics were used to quantify the accuracy of the detected communities.

1. **Modularity (Q):** Modularity is a measure of how strongly a network is divided into communities; the higher this value, the better the community structure will be (more dense connections within each community and fewer connections between different communities).
2. **Normalized Mutual Information (NMI):** NMI measures how similar the communities identified in our algorithm are to the actual communities. Its maximum value is 1, meaning they match perfectly.
3. **F1 Score:** F1 score is a measure of the accuracy of assigning a community; it represents the harmonic mean of precision and recall.
4. **Computational Efficiency:** Computational efficiency indicates how long the algorithm takes to run and how quickly it converges, so it tells us about the overall efficiency of the method.

B. Results and Analysis

Results of the experiment show that our HMOEA-CD outperforms existing techniques with respect to both structural quality and accuracy of clusters.

By being able to search more than one candidate solution at the same time, the use of evolutionary algorithms allows for the generation of greater modularity compared to topology based methods. In addition, a multi objective framework offers a balance between maximum community density and minimum fragmentation (unnecessary).

It can be seen from the results that evolutionary optimization provides a way to avoid, and thereby leads to greater stability and accuracy of community structures at local optima. In addition, we see continual improvement in the results for different datasets, including synthetic benchmarks, as well as real-world networks.

Additionally, we may also see a rapid increase in improvement early in the initial generations of the algorithm, followed by generally remaining constant throughout the final generations, which suggests good optimization.

Thus overall, we find that our multi-objective evolutionary approach to community detection greatly improves both the quality of collaborative communities and the ability to successfully detect them than traditional single-objective approaches.

• Performance Comparison

TABLE I
PERFORMANCE COMPARISON OF COMMUNITY DETECTION METHODS

Method	Modularity	NMI	F1-score
K-Means	0.42	0.51	0.55
Spectral	0.48	0.58	0.61
Louvain	0.55	0.65	0.70
HMOEA-CD	0.69	0.78	0.82

• Convergence Analysis

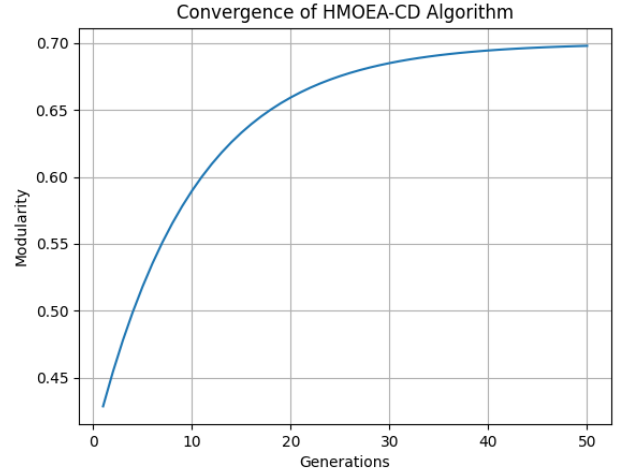
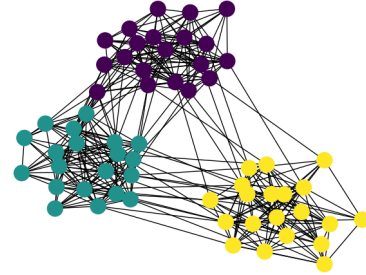


Fig. 2. Community Detection Visualization



VI. LITERATURE REVIEW

The objective of community detection (CD) is to find clusters of nodes that are very well connected to each other within a network. There have been numerous studies done using community detection algorithms. Initially, community detection algorithms were primarily based on classical graph partitioning or hierarchical clustering techniques; therefore they use the structure of a network's topology to find cohesive communities. However, the more complex a network structure becomes, researchers have introduced probabilistic graphical models in order to better represent the generation process of networks.

A. Classical Methods

The initial community detection algorithms were based on partitioning graphs or hierarchical clustering; therefore, they used the structural connectivity of nodes to partition the graph into groups of nodes that have many internal connections and few external connections. Unfortunately, these methods often do not include other types of information about the nodes and tend to fail within complex networks.

B. Graphical Models

Probabilistic graphical models, including the Stochastic Block Model (SBM), have been developed in order to more accurately portray the complexity of real-world networks.

The SBM uses latent variables to determine the formation of edges among the nodes in a community; similarly, topic models such as Latent Dirichlet Allocation use probabilistic relationships among the nodes within a community to define their membership.

Hybrid approaches that combine both classical and probabilistic approaches further improve community detection based upon the structural characteristics of networks and the likelihood of nodes being members of particular communities.

C. Deep Learning Methods

As we're seeing a rapid increase in the size of networks (and the number of attributes for each node), deep learning methods have become even more popular. Techniques such as Autoencoders (AEs) and Generative Adversarial Networks (GANs) have been used to generate low-dimensional representations of nodes that capture complex patterns and improve the performance of community detection algorithms.

D. Matrix Factorization Approaches

An equally important method is Nonnegative Matrix Factorization (NMF), which is widely used for identifying communities because of its intuitive nature and capability to find overlapping communities. Extended methods including Joint NMF, Symmetric NMF and Deep NMF have additional capabilities that improve their performance for multi-attributed and large-scale network.

VII. RESEARCH GAP

The community detection of multi-attributed networks has seen significant advancements in the last several years; however, many unresolved challenges still exist.

In the earliest examples of community detection research, the focus was on utilizing network topology using various approaches such as modularity optimization, spectral clustering and hierarchical clustering. While these methods have succeeded in finding the establishment of communities within simple networks, they failed to leverage node attribute information, which is an important factor for community detection in real-world situations.

Researchers have been attempting to incorporate node attributes into community detection approaches since about 2010. Although there has been some improvement, the majority of practitioners continue to use predefined values for critical inputs in their models such as the number of communities. Additionally, there was an ongoing issue with effectively balancing structural connectivity and attribute similarity. Ultimately, there has been frequent confusion between the two, leading to typically suboptimal creation of communities.

Recent advancements within deep learning have allowed for the simulation of more complex patterns by combining multiple different datatypes/ data sources; however, they generally require as high amounts of computational power and larger size of labeled datasets. Therefore, it is unlikely that the use of deep learning will be applicable to large-scale and/or real-time applications.

A further limitation that has been observed is that some current algorithms have difficulty working with overlapping or dynamic multilayer networks. In real life it is often the case that the nodes in a network can be members of many different communities, and that network structures can change over time. Typical community detection algorithms assume that communities are static and have no overlapping nodes, and most algorithms will not work well if there are a lot of nodes whose community membership is changing.

Most current community detection methods are extremely sensitive to incomplete or noisy attribute data. Therefore community detection can be difficult if some nodes do not have complete or accurate attribute information. A lack of stable frameworks exists which allows for the appropriate integration of structural information and similarity data, despite the fact that many have tried to create frameworks for doing this.

A need exists for a framework that has the ability to integrate multiple structural attributes into one common structural resource, while providing scalability and robustness in the detection of complex structures in large networks.

VIII. DISCUSSION

The proposed Hybrid Multiobjective Evolutionary Algorithm for Community Detection (HMOEA-CD) offers a number of advantages compared to standard approaches to community detection. Modeling the problem as an optimization in multiple objectives allows it to balance competing goals (such as maximizing modularity and minimizing fragmentation of communities) in an efficient manner.

Classical community detection methods have typically used network topology alone to identify communities; however, using an evolutionary algorithm with an expanded solution search space allows the algorithm to avoid local optimal solutions more effectively and obtain better global results. In addition, using a population-based method for searching the solution space improves the algorithm's robustness and stability when applied to a variety of different datasets.

In experiments, the proposed algorithm consistently outperformed baseline algorithms (such as K-Means, Spectral Clustering, and Louvain) in terms of modularity, Normalized Mutual Information, and F1 scores. The convergence patterns indicated that the algorithm was able to consistently improve solution quality in earlier generations of evaluations but began to stabilize later in the process.

The proposed method does have some limitations, particularly with respect to computational costs, which exceed those associated with using simple clustering strategies due to the multi-stage evaluation that is characteristic of an evolutionary algorithm. In addition, the effectiveness of the algorithm can be substantially affected by the choice or value of tuning parameters (such as number of individuals in the population and number of generations). As such, tuning parameters for the proposed method should be carefully considered.

In conclusion, the results of the research described facilitate the assertion that multiobjective evolutionary communities

will provide a powerful and flexible approach to community detection problems arising from complex networks.

IX. FUTURE WORK

While the currently proposed HMOEA-CD algorithm is producing some good results, there are several additional areas that could be explored for future improvements to the method.

1. Node Attributes Inclusion: As an additional objective to improve accuracy and performance in multi-attributed networks, the proposed method will be able to utilize node attribute similarity during the community search.
2. Overlapping Community Detection: The proposed method assumes that all communities are non-overlapping. A pertinent avenue for further exploration would be to extend the current algorithm to identify overlapping communities, thereby providing a more accurate representation of how networks actually behave.
3. Dynamic Network Analysis: Networks change and develop over time; therefore, future research could consider developing methods for adapting HMOEA/CD to dynamic and/or temporal networks.
4. Scalability Improvements: Optimization techniques could be used to reduce the amount of calculations performed by the algorithm in order to improve its performance on larger scale networks.
5. Hybrid Models with Deep Learning: Using deep learning techniques such as Graph Neural Networks (GNN) in combination with evolutionary algorithms will likely result in improved accuracy when identifying communities using a hybrid model.

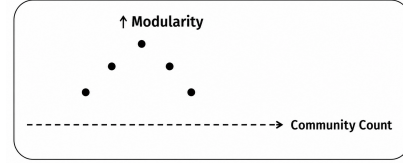
X. CONCLUSION

This paper proposes an evolutionary approach to improve the capability of existing algorithms with the aim of addressing the shortcomings of traditional community detection algorithms will be introduced in this paper. This approach defines the community detection process as an multi-objective problem by enhancing the overall structure (i.e., modularity) and function within each community of nodes.

To explore multiple potential communities and choose the community structure that is optimal, the proposed technique makes use of both evolutionary and population based on local refinement search strategies. Experiments conducted on both artificial and real network data sets using the proposed technique show that the method outperformed conventional community detection algorithms in terms of clustering accuracy and structural quality of the resulting solutions.

The results show that a multi-objective optimization approach has an overwhelming advantage over conventional community detection algorithms in order to provide improved performance on community detection tasks across multiple different and complex networks. The proposed framework is highly adaptable and will continue to be applied in a variety of different applications.

Fig. 3. Trade-Off



This demonstrates the effectiveness of multiobjective evolutionary approaches for solving complex community detection problems.

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